

Sample Paper 9 — Class 5 Mathematics Olympiad (Level-2 Difficulty Step-Up)

Pattern: Same 50 questions / 60 marks / 60 minutes / no negative marking as Papers 1–8, but noticeably harder throughout — closer to SOF IMO **Level 2**.

Section 1 — Logical Reasoning (Q1–Q15, 1 mark each)

- Q1.** Find the odd one out: A) Hockey B) Football C) Cricket D) Chess
- Q2.** Doctor is to Patient as Teacher is to _____. A) Student B) School C) Lesson D) Book
- Q3.** If each letter of a word is moved 1 step forward in the alphabet, CAT becomes DBU. Using the same rule, what does HORSE become? A) IPSTF B) IPSTE C) IQSTF D) JPSTF
- Q4.** This series follows two interleaved patterns — the 1st, 3rd, 5th, 7th terms, and the 2nd, 4th, 6th, 8th terms. Find the missing (8th) term: 5, 100, 10, 93, 15, 86, 20, ____ A) 77 B) 78 C) 79 D) 80
- Q5.** Five friends — Rhea, Sameer, Tanvi, Umesh, Vikas — sit in a row. Tanvi sits exactly in the middle. Vikas sits at the left end. Umesh sits immediately to the right of Tanvi. Sameer sits between Vikas and Tanvi. Find the seating order, left to right. A) Vikas, Sameer, Tanvi, Umesh, Rhea B) Vikas, Tanvi, Sameer, Umesh, Rhea C) Sameer, Vikas, Tanvi, Umesh, Rhea D) Vikas, Sameer, Umesh, Tanvi, Rhea
- Q6.** Anita walks 16 m North, then 9 m further North, then 7 m South. How far is she from her starting point, and in which direction? A) 16 m North B) 18 m North C) 20 m North D) 25 m North
- Q7.** If the 11th day of a month is a Thursday, what day of the week is the 30th of the same month? A) Monday B) Tuesday C) Wednesday D) Sunday
- Q8.** Find the odd one out: A) 21 B) 28 C) 42 D) 45 (Hint: three of these are multiples of the same number.)
- Q9.** Find the next number: 5, 10, 40, 200, ____ A) 1,000 B) 1,100 C) 1,200 D) 1,300
- Q10.** In a queue, Farah is 12th from the front and 18th from the back. If 4 people leave from the back end, what will be her new position from the back? A) 12th B) 13th C) 14th D) 15th
- Q11.** Find the odd one out: A) 13 B) 29 C) 39 D) 43
- Q12.** In a certain code, BOOK is written as ERRN (each letter moved 3 steps forward). How is PENCIL written in the same code? A) SHQFLO B) SHQFLP C) SHRFLO D) THQFLO
- Q13.** Find the odd one out: A) Pentagon B) Hexagon C) Octagon D) Cylinder

Q14. Four friends ran a race. Priya finished before Rina but after Sana. Tina finished after Rina. Who finished last? A) Priya B) Rina C) Sana D) Tina

Q15. Study the pattern: 3, 4, 6, 9, 13, _____. Find the next number. A) 17 B) 18 C) 19 D) 20

Section 2 — Mathematical Reasoning (Q16–Q35, 1 mark each)

Q16. $4,89,675 + 3,78,456 = ?$ A) 8,68,131 B) 8,67,131 C) 8,68,031 D) 8,69,131

Q17. $7,60,000 - 4,89,675 = ?$ A) 2,70,325 B) 2,69,325 C) 2,71,325 D) 2,70,225

Q18. $6,543 \times 14 = ?$ A) 91,602 B) 91,502 C) 91,702 D) 92,602

Q19. $97,776 \div 18 = ?$ A) 5,432 B) 5,422 C) 5,442 D) 5,332

Q20. Find the HCF of 108 and 144. A) 18 B) 24 C) 36 D) 72

Q21. Find the LCM of 16 and 24. A) 24 B) 36 C) 48 D) 96

Q22. Which of these fractions is the greatest: $\frac{8}{9}$, $\frac{5}{6}$, $\frac{13}{15}$, $\frac{10}{11}$? A) $\frac{8}{9}$ B) $\frac{5}{6}$ C) $\frac{13}{15}$ D) $\frac{10}{11}$

Q23. $6 \text{ km } 250 \text{ m} + 4 \text{ km } 900 \text{ m} = ?$ A) 10 km 150 m B) 11 km 50 m C) 11 km 150 m D) 10 km 950 m

Q24. A rectangular field is 50 m long and 34 m wide. Find its area. A) 1,680 sq m B) 1,690 sq m C) 1,700 sq m D) 1,710 sq m

Q25. A regular nonagon (9 sides) has a perimeter of 108 cm. Find the length of each side. A) 10 cm B) 11 cm C) 12 cm D) 13 cm

Q26. The cost price of an item is ₹1,800 and the selling price is ₹2,250. Find the profit percentage. A) 20% B) 22.5% C) 25% D) 27.5%

Q27. A sum of money doubles itself in 10 years at simple interest. Find the rate of interest per annum. A) 8% B) 10% C) 12% D) 15%

Q28. ₹600 is divided among three people in the ratio 3:4:5. Find the largest share. A) ₹225 B) ₹240 C) ₹250 D) ₹260

Q29. Convert 0.45 to a fraction in simplest form. A) $\frac{9}{20}$ B) $\frac{4}{9}$ C) $\frac{9}{25}$ D) $\frac{5}{11}$

Q30. $7 \text{ kg } 300 \text{ g} + 5 \text{ kg } 650 \text{ g} + 4 \text{ kg } 550 \text{ g} = ?$ A) 16 kg 500 g B) 17 kg C) 17 kg 500 g D) 18 kg

Q31. How many seconds are there in 5.5 hours? A) 18,900 B) 19,400 C) 19,800 D) 20,200

Q32. Find the perimeter of an equilateral triangle with side 38 cm. A) 76 cm B) 104 cm C) 114 cm D) 38 cm

Q33. Find the area of a rectangle with length 55 cm and breadth 36 cm. A) 1,960 sq cm B) 1,970 sq cm C) 1,980 sq cm D) 1,990 sq cm

Q34. How many faces does a triangular pyramid (tetrahedron) have? A) 3 B) 4 C) 5 D) 6

Q35. Find the number of diagonals in a nonagon (9 sides). A) 24 B) 25 C) 26 D) 27

Section 3 — Everyday Mathematics (Q36–Q45, 1 mark each)

Q36. A school collected ₹3,15,000 from 7 classes equally. How much did each class contribute? A) ₹44,000 B) ₹44,500 C) ₹45,000 D) ₹45,500

Q37. A factory makes 3,600 bottles in 10 hours, at a steady rate. How many bottles does it make in 7 hours? A) 2,420 B) 2,470 C) 2,520 D) 2,570

Q38. A tank holds 6,000 L and is filled by a pipe at 200 L per minute. How long will it take to fill the tank completely? A) 28 min B) 29 min C) 30 min D) 32 min

Q39. A bus leaves city A at 6:45 a.m., stops for 25 minutes on the way, and reaches city B at 12:00 noon. How much time did the bus spend actually travelling (excluding the stop)? A) 4 h 30 min B) 4 h 40 min C) 4 h 50 min D) 5 h

Q40. A chemist mixes 6 L 500 mL of one liquid with 3 L 500 mL of another, then pours the mixture equally into 5 bottles. How much does each bottle get? A) 1 L 800 mL B) 1 L 900 mL C) 2 L D) 2 L 100 mL

Q41. Rohit buys 6 notebooks at ₹42 each and gets a discount of ₹22 on the total bill. How much does he pay? A) ₹220 B) ₹225 C) ₹230 D) ₹235

Q42. A cistern with 2,100 L of water is used for irrigation at the rate of $\frac{1}{6}$ of its original capacity every day. How much water remains after 3 days? A) 1,000 L B) 1,050 L C) 1,100 L D) 1,150 L

Q43. A delivery van carries 4 boxes weighing 23 kg 400 g, 19 kg 600 g, 16 kg 250 g, and 12 kg 750 g. Find the total weight carried. A) 71 kg B) 71 kg 500 g C) 72 kg D) 72 kg 500 g

Q44. 2,352 chairs are arranged equally into rows of 42 chairs each. How many rows will there be? A) 52 B) 54 C) 56 D) 58

Q45. A trader bought 85 kg of wheat for ₹3,400 and sold all of it at ₹46 per kg. Find his total profit. A) ₹470 B) ₹490 C) ₹510 D) ₹530

Section 4 — Achievers Section (Q46–Q50, 3 marks each)

Q46. A number is first multiplied by 8, then 32 is subtracted from the result, giving 144. Find the original number. A) 20 B) 21 C) 22 D) 23

Q47. The sum of three numbers is 225. The second number is twice the first, and the third is 25 more than the second. Find the first number. A) 36 B) 38 C) 40 D) 42

Q48. Priyanka had a certain amount of money. She spent $\frac{1}{3}$ of it on books, then $\frac{1}{5}$ of what remained on stationery, and was left with ₹1,200. How much money did she have at first? A) ₹2,100

B) ₹2,175 C) ₹2,250 D) ₹2,325

Q49. In a school, the number of girls is 9 more than twice the number of boys. There are 354 students in total, and 115 of them are boys. Find the number of girls. A) 233 B) 236 C) 239 D) 242

Q50. A sequence starts with 4 and 7, and each later term is the sum of the two terms before it: 4, 7, 11, 18, 29, _____. Find the next term. A) 41 B) 44 C) 47 D) 50

Answer key: `sample-paper-09-answer-key.md`
